

\*\*\*\*\*  
\*\*\*\*\*

labprogram1:–

- 1. Create a form with the elements of Textboxes, Radio buttons, Checkboxes, and so on.  
Write JavaScript code to validate the format in email, and mobile number in 10 characters, If a textbox has been left empty, popup an alert indicating when email, mobile number and textbox has been left empty.**

```
<html>

<head>

<title>Form Validation Without Regex</title>

<script>

function validateForm() {

    let email = document.getElementById("email").value;

    let mobile = document.getElementById("mobile").value;

    let name = document.getElementById("name").value;

    // Basic validation without regular expressions

    if (name.trim() === "") {

        alert("Name field cannot be empty.");

        return false;

    }

    if (email.indexOf("@") === -1 || email.indexOf(".") === -1 || email.length < 5) {

        alert("Please enter a valid email address.");

        return false;

    }

    if (mobile.length !== 10 || isNaN(mobile)) {

        alert("Mobile number must be exactly 10 digits.");

    }

}
```

```
        return false;
    }

    alert("Form submitted successfully!");

    return true;
}

</script>

</head>

<body>

    <form onsubmit="return validateForm()">

        <label>Name: <input type="text" id="name"></label><br></br>

        <label>Email: <input type="text" id="email"></label><br></br>

        <label>Mobile: <input type="text" id="mobile"></label><br></br>

        <input type="submit" value="Submit">

    </form>

</body>

</html>
```

\*\*\*\*\*

Labprogram2:--

**2.Develop an HTML Form, which accepts any Mathematical expression. Write JavaScript code to evaluate the expression and Display the result.**

```
<html>

<title> Arithmetic expression Evaluation </title>

<script type="text/javascript">

    function evaluateExpression() {

        var enteredExpr = document.getElementById("expr").value;

        document.getElementById("result").value = eval(enteredExpr);
```

```

        alert("expression is evaluated correct");

    }

    function clearExpression() {

        document.getElementById("expr").value="";

        document.getElementById("result").value="";

        alert("invalid expression ");a

    }

</script>

</head>

<body bgcolor=cyan>

    <h1 align=center> Evaluating Arithmetic Expression</h1>

    <hr>

    <form name="myform">

        <b>Enter any valid Expression : <input type=text id=expr></b>

        <b><input type=button value="Evaluate" onclick="evaluateExpression()"></b>

        <b><input type=button value="clear" onclick="clearExpression()"></b>

        <b>Result of the expression : <input type=text id=result></b>

    </form>

</body>

</html>

```

\*\*\*\*\*

Lab program3:—Create a page with dynamic effects. Write the code to include layers and basic animation.

```

<html>

<head>

    <title> Basic Animation </title>

    <style>

        #layer1 {

```



```
position: absolute;

top: 50px;

left: 150px;

}
```

```
#layer2 {

position: absolute;

top: 50px;

left: 250px;

}
```

```
#layer3 {

position: absolute;

top: 50px;

left: 350px;

}
```

```
</style>
```

```
<script type="text/javascript"> function moveImage(layer) {

    var top = window.prompt("Enter Top value");

    var left = window.prompt("Enter Left Value");

    document.getElementById(layer).style.top = top + 'px';

    document.getElementById(layer).style.left = left + 'px';

}
```

```
</script>
```

```
</head>
```

```
<body bgcolor=cyan>
```

```
<div id="layer1"></div>
```

```
<div id="layer2"></div>
```

```
<div id="layer3"></div>
```

</body>

</html>

.....

Lab program 4:–Write a JavaScript code to find the sum of N natural Numbers. (Use user defined function)

<html>

<head>

<title> Sum of Natural Numbers </title>

<script type="text/javascript">

function sum() {

var num = window.prompt("Enter the value of N");

var n = parseInt(num);

var sum = (n \* (n + 1)) / 2;

window.alert("Sum of First " + n + " Natural numbers is: " + sum);

}

</script>

</head>

<body bgcolor=cyan>

<h1 align=center> Finding Sum of N Natural Numbers </h1>

<hr>

<form align=center> <input type="button" value="Click Here " onclick="sum()" /> </form>

</body>

</html>

\*\*\*\*\*

Lab Program 5. Write a JavaScript code block using arrays and generate the current date in words, this should include the day, month and year.

<html>

<head>

<title>Display Date </title>

<script type="text/javascript">

```
function display() {

    var dateObj = new Date();

    var currDates = dateObj.getDate();

    var month = dateObj.getMonth();

    var currYear = dateObj.getFullYear();

    var year = "Two Thousand and Twenty Five";

    var days = ["First", "Second", "Third", "Fourth", "Fifth", "Sixth", "Seventh", "Eigth", "Ninth", "Tenth",
"Eleventh", "Twelfth", "Thirteenth", "Fourteenth", "fifteenth", "Sixteenth", "Seventeenth", "Eighteenth",
"Nineteenth", "Twentyeth", "Twenty First", "TwentySecond", "Twenty Third", "Twenty Fourth", "Twenty Fifth",
"Twenty Sixth", "TwentySeventh", "Twenty Nine", "Thirtyeth", "Thirty First"];

    var months = ["January", "Febraury", "March", "April", "May", "June", "July", "August", "September",
"October", "November", "December"];

    if (currYear == 2025)

        alert("Today date is :: " + days[currDates - 1] + " " + months[month] + " " + year);

    else

        alert("Today date is not to the current year");

}

</script>

</head>

<body bgcolor=cyan>

    <form >

        <input type=button value="Click Here" onClick="display()" />

    </form >

</body>

</html>
```

Lab program 6:– **Create a form for Student information. Write JavaScript code to find Total, Average, Result and Grade.**

```
1.
2. <html>
3. <head>
4.   <title>Student Data Example</title>
5.   <script type="text/javascript">
6.     function showResult(){
```



```
7.     varname = document.getElementById("name").value;
8.     varcls = document.getElementById("class").value;
9.     varmarks1 = parseInt(document.getElementById("sub1").value);
10.    varmarks2 = parseInt(document.getElementById("sub2").value);
11.    varmarks3 = parseInt(document.getElementById("sub3").value);
12.    var total = marks1 + marks2 + marks3;
13.    var avg = total / 3;
14.    var grade, result;
15.    if (avg >= 60) {
16.        grade = "A";
17.        result = "First Class";
18.    }
19.    else if (avg < 60 && avg >= 50) {
20.        grade = "B";
21.        result = "Second Class";
22.    }
23.    else if (avg < 50 && avg >= 40) {
24.        grade = "C";
25.        result = "Third Class";
26.    }
27.    else {
28.        grade = "D";
29.        result = "Fail";
30.    }
31.    document.write("<h2>Results</h2>");
32.    document.write("<b>Name: " + name + "</b><br />");
33.    document.write("<b>Class: " + cls + "</b><br />");
34.    document.write("<b>Total Marks: " + total + "</b><br />");
35.    document.write("<b>Average: " + avg + "</b><br />");
36.    document.write("<b>Grade: " + grade + "</b><br />");
37.    document.write("<b>Result: " + result + "</b><br />");
38.    }
39. </script>
40. </head>
41. <body>
42.     <form>
43.         <tr>
44.             <td>Student Name:</td><br/>
45.             <td><input type="text" id="name" ></td><br/>
46.             <td>Student Class:</td><br/>
47.             <td><input type="text" id="class" ></td><br/>
48.         </tr>
49.         <tr>
50.             <td>Student Marks1:</td><br/>
51.             <td><input type="number" id="sub1" ></td><br/>
52.             <td>Student Marks2:</td><br/>
53.             <td><input type="number" id="sub2" ></td><br/>
54.             <td>Student Marks3:</td><br/>
55.             <td><input type="number" id="sub3" ></td><br/>
56.         </tr>
57.     </table>
```

```
58.     <input type="button" value="View Results" onClick="showResult()"><br/>
59.   </form></body></html>
60.
```

Lab Program 7. Create a form for Employee information. Write JavaScript code to find DA, HRA, PF, TAX, Gross pay, Deduction and Net pay.

```
<html>
```

```
<head>
```

```
  <title>Employee Salary Calculator</title>
```

```
</head>
```

```
<body>
```

```
  <h2>Employee Salary Calculator</h2>
```

```
  <label>Basic Salary:
```

```
    <input type="number" id="basic">
```

```
</label><br>
```

```
  <button onclick="calculate()">Calculate Salary</button>
```

```
  <h3>Results:</h3>
```

```
  <p id="result"></p>
```

```
<script>
```

```
  function calculate() {
```

```
    let basic = parseFloat(document.getElementById("basic").value);
```

```
    if (isNaN(basic) || basic <= 0) {
```

```
      document.getElementById("result").innerHTML = "Please enter a valid basic salary.";
```



```
return;

}

// Calculations

let da = basic * 0.20; // 20%

let hra = basic * 0.25; // 25%

let pf = basic * 0.12; // 12%

let tax = basic * 0.10; // 10%


let gross = basic + da + hra;

let deduction = pf + tax;

let net = gross - deduction;


document.getElementById("result").innerHTML =

"Gross Pay: ₹ " + gross.toFixed(2) + "<br>" +

"Deduction: ₹ " + deduction.toFixed(2) + "<br>" +

"Net Pay: ₹ " + net.toFixed(2);

}

</script>

</body>

</html>
```

### Lab program 8:–

Write a program in PHP to change background colour based on day of the week using if else if statements and using arrays.

```
<!DOCTYPE html>
<html>
<head>
<title>Day-Based Background Color</title>
<style>
body{
```

```
font-family: Arial, sans-serif;
text-align: center;
}
</style>
</head>
<body>

<?php
// Array to store day names and corresponding colors
$days = array(
    "Sunday" => "orange",
    "Monday" => "white",
    "Tuesday" => "red",
    "Wednesday" => "green",
    "Thursday" => "yellow",
    "Friday" => "pink",
    "Saturday" => "gray"
);

// Get the current day of the week
$today = date("l");

// Set the background color based on the day
echo "<body style='background-color: " . $days[$today] . "';>";

// Display a message with the day and color
echo "<h1>Today is " . $today . "</h1>";
echo "<p>The background color is " . $days[$today] . "</p>";

echo "</body>";
?>

</html>
```

\*\*\*\*\*

\*\*\*\*\*lab program 9 :--

Write a simple program in PHP for i) generating Prime number ii) generate

Fibonacci series.

```
<?php

function isPrime($num) {

    if ($num < 2) return false;

    for ($i = 2; $i <= $num/2; $i++) {
```

```
        if ($num % $i == 0) return false;

    }

    return true;

}
```

```
$limit = 10; // Change this to any desired limit
```

```
echo "Prime numbers up to $limit:\n";
```

```
for ($i = 2; $i <= $limit; $i++) {
```

```
    if (isPrime($i)) {
```

```
        echo "$i ";
```

```
    }
```

```
}
```

```
?>
```

b) <?php

```
function fibonacci($n) {
```

```
    $a = 0;
```

```
    $b = 1;
```

```
    echo "Fibonacci series up to $n terms:\n$a $b ";
```

```
    for ($i = 2; $i < $n; $i++) {
```

```
        $next = $a + $b;
```

```
        echo "$next ";
```

```
        $a = $b;
```

```
        $b = $next;
```

```
    }
```

```
}
```

```
$terms = 10; // Change this to generate more terms
```



```
fibonacci($terms);
```

```
?>
```

### Lab program 10:--

Write a PHP program to remove duplicates from a sorted list

```
<?php
```

```
function removeDuplicates($array){
```

```
    $result=array_values(array_unique($array));
```

```
    return $result;
```

```
}
```

```
$sortedList=[1,1,2,2,3,3,4,5,5];
```

```
$uniqueList=removeDuplicates($sortedList);
```

```
echo"Original List:";
```

```
print_r($sortedList);
```

```
echo"<br> Unique List:";
```

```
print_r($uniqueList);
```

```
?>
```

Labprogram1 1:– Write a PHP Script to print the following pattern on the Screen:

```
*****
```

```
****
```

```
***
```

```
**
```

```
*
```

```
<?php
```

```
for ($i = 5; $i >= 1; $i--) {
```

```
    for ($j = 0; $j < $i; $j++) {
```

```
        echo "*";
```

```
}  
  
echo "\n";  
  
}  
  
?>
```

Lab program 12. Write a simple program in PHP for Searching of data by different criteria

```
<?php  
  
$data=[  
  
    ['id'=>1,'name'=>'Srikanth','age'=>30],  
  
    ['id'=>2,'name'=>'Srinath','age'=>35],  
  
    ['id'=>3,'name'=>'Srinivas','age'=>50],  
  
    ['id'=>4,'name'=>'Smayan','age'=>45],  
  
    ['id'=>4,'name'=>'Saatvik','age'=>50],  
  
];  
  
function searchByCriteria($data,$criteria){  
  
    $results=[];  
  
    foreach($data as $entry){  
  
        $match=true;  
  
        foreach($criteria as $key=>$value){  
  
            if(!isset($entry[$key])||$entry[$key]!=$value){  
  
                $match=false;  
  
                break;  
  
            }  
  
        }  
  
        if($match){  
  
            $results[]=$entry;  
  
        }  
  
    }  
  
}
```

```
}

return $results;

}

$criteria=['age'=>50];

$searchResults=searchByCriteria($data,$criteria);

echo "Search Results:<br>";

//print_r($searchResults);

foreach ($searchResults as $result) {

    echo "ID: " . $result['id'] . "<br>";

    echo "Name: " . $result['name'] . "<br>";

    echo "Age: " . $result['age'] . "<br>";

}

?>
```

### LAB PROGRAM 13. Write a function in PHP to generate captcha code

```
<?php

function generateCaptcha($length = 6) {

    // Define possible characters for the captcha

    $characters = 'ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789';

    $captcha = "";

    // Generate a random string of given length

    for ($i = 0; $i < $length; $i++) {

        $captcha .= $characters[rand(0, strlen($characters) - 1)];

    }

    return $captcha;

}
```



```
// Usage example

$captchaCode = generateCaptcha();

echo "generated captcha code is :";

echo ($captchaCode);

?>
```

**LABPROGRAM 14:-- Write a Program to store and read image from Database.**

```
<!DOCTYPE html>

<html>

    <body>

        <form method="post" enctype="multipart/form-data">

            <input type="file" name="image">

            <input type="submit" value="Upload">

        </form>

        <?php

            $conn = new mysqli("localhost", "root", "root", "images");

            if (isset($_FILES['image'])) {

                $name = $_FILES['image']['name'];

                $tmp = $_FILES['image']['tmp_name'];

                $path = "uploads/" . $name;

                if (!is_dir("uploads")) mkdir("uploads");

                move_uploaded_file($tmp, $path);

                $conn->query("INSERT INTO image_paths (name, path) VALUES ('$name', '$path')");

                echo "✅ Image uploaded successfully!";

            }

        ?>

    </body>
```

```
</html>
```

LAB PROGRAM 15:– Write a program in PHP to read and write file using form control.

```
<html>
```

```
<head>
```

```
    <title>File Read/Write Example</title>
```

```
</head>
```

```
<body>
```

```
    <h2>Write to File</h2>
```

```
    <form method="post">
```

```
        <textarea name="content" rows="5" cols="40" placeholder="Type something..."></textarea><br><br>
```

```
        <input type="submit" name="submit" value="Save to File">
```

```
    </form>
```

```
<?php
```

```
$filename = "data.txt";
```

```
// Write to file
```

```
if (isset($_POST['submit'])) {
```

```
    $text = $_POST['content'];
```

```
    file_put_contents($filename, $text); // Overwrites existing content
```

```
    echo "<p><strong>Content saved!</strong></p>";
```

```
}
```

```
// Read from file
```

```
if (file_exists($filename)) {
```

```
    echo "<h2>File Contents:</h2>";
```



```
$contents = file_get_contents($filename);

echo "<pre>" . htmlspecialchars($contents) . "</pre>";

}

?>

</body>

</html>
```

**LABPROGRAM 16:-- Write a program in PHP to add, update and delete using student database**

```
<?php

$conn = mysqli_connect("localhost","root","root","studentdb") or die("DB Error");

// Add or Update

if(isset($_POST['save'])) {

    $id=$_POST['id'];

    $name=$_POST['name'];

    $age=$_POST['age'];

    $course=$_POST['course'];

    $q = $id ? "UPDATE students SET name='$name',age=$age,course='$course' WHERE id=$id"

        : "INSERT INTO students(name,age,course) VALUES('$name',$age,'$course)";

    mysqli_query($conn,$q);

}

// Delete

if(isset($_POST['delete'])) { mysqli_query($conn,"DELETE FROM students WHERE id=$_POST[id]"); }

// Prepare Edit

$edit=['id'=>','name'=>','age'=>','course'=>];
```

```
if(isset($_POST['edit'])){ $r=mysqli_query($conn,"SELECT * FROM students WHERE id=$_POST[id]");
    $edit=mysqli_fetch_assoc($r);}

    ?>
```

```
<form method="POST">
```

```
<input type="hidden" name="id" value="<?=$edit['id']?>">
```

```
Name:<input name="name" value="<?=$edit['name']?>" required>
```

```
Age:<input type="number" name="age" value="<?=$edit['age']?>" required>
```

```
Course:<input name="course" value="<?=$edit['course']?>" required>
```

```
<input type="submit" name="save" value="Save">
```

```
<?php if($edit['id']): ?><input type="submit" name="delete" value="Delete"><?php endif; ?>
```

```
</form>
```

```
<table border=1 cellpadding=5>
```

```
<tr><th>ID</th><th>Name</th><th>Age</th><th>Course</th><th>Edit</th></tr>
```

```
<?php $res=mysqli_query($conn,"SELECT * FROM students");
```

```
while($row=mysqli_fetch_assoc($res)){
```

```
    echo "<tr><td>$row[id]</td><td>$row[name]</td><td>$row[age]</td><td>$row[course]</td><
```

```
<td><form method='POST' style='margin:0;'><input type='hidden' name='id' value='$row[id]'><input
type='submit' name='edit' value='Edit'></form></td></tr>";} ?>
```

```
</table>
```

## LAB program 17:– Write a program in PHP to Validate Input

```
<!DOCTYPE html>

<html>

<head>

    <title>Input Validation</title>

</head>

<body>

    <form method="post">

        Name:<input type="text" name="name"><br><br>

        Email:<input type="text" name="email"><br><br>

        <input type="submit" name="submit" value="Submit">

    </form>

</body>

</html>

<?php

if($_SERVER["REQUEST_METHOD"]=="POST"){

    $name=$_POST['name'];

    $email=$_POST['email'];

    if(empty($name)){

        echo "Name is required.<br><br>";

    } else{

        echo "Name: ". $name . "<br><br>";

    }

    if(!empty($email)){

        echo "Email is given.<br><br>";

    }

    else{
```



```
        echo "Email:is to be given " . $email . "<br><br>";

    }

}

?>
```

### LAB PROGRAM 18:– Write a program in PHP for setting and retrieving a cookie

```
<?php

// Set a cookie named "user" with value "Vanishree" that expires in 1 hour

setcookie("user", "Vanishree", time() + 3600, "/"); // "/" means available across the whole site

?>
```

```
<!DOCTYPE html>

<html>

<head>

    <title>Cookie Example</title>

</head>

<body>

    <h2>Cookie Demo</h2>
```

```


    <?php

        // Check if the cookie is set

        if (isset($_COOKIE["user"])) {

            echo "Welcome back, " . $_COOKIE["user"] . "!";

        } else {

            echo "Cookie not set yet. Refresh the page after setting.";

        }

    ?>
```

```
</body>
```

```
</html>
```

LAB PROGRAM 19:– Write a PHP program to Create a simple webpage of a college

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Welcome to ABC College</title>
```

```
    <style>
```

```
        body {
```

```
            font-family: Arial, sans-serif;
```

```
            background-color: #eef;
```

```
            margin: 20px;
```

```
        }
```

```
        section {
```

```
            background-color: #fff;
```

```
            padding: 15px;
```

```
            margin-top: 10px;
```

```
            border-radius: 5px;
```

```
        }
```

```
        h2 {
```

```
            color: #004080;
```

```
        }
```

```
    </style>
```

```
</head>
```

```
<body>
```

```
<header>

    <h1>ABC College</h1>

    <p>Empowering Education Since 1990</p>

</header>
```

```
<section>

    <h2>About Us</h2>

    <p>ABC College is a premier institution offering quality education in Science, Commerce, and Arts streams.</p>

</section>
```

```
<section>

    <h2>Departments</h2>

    <?php

$departments = ["Computer Science", "Physics", "Commerce", "English Literature", "Mathematics"];

echo "<ul>";

foreach ($departments as $dept) {

    echo "<li>$dept</li>";

}

echo "</ul>";

?>

</section>
```

```
<section>

    <h2>Contact Information</h2>

    <?php

$email = "info@abccollege.edu";
```



```
$phone = "+91-9876543210";

$address = "123 College Road, Bengaluru, Karnataka";


echo "<p>Email: $email</p>";

echo "<p>Phone: $phone</p>";

echo "<p>Address: $address</p>";

?>

</section>


</body>

</html>
```

LAB PROGRAM20:-- Write a program in PHP for exception handling for

i) divide by zero

ii) checking date format.

```
<!DOCTYPE html>

<html>

<head>

    <title>Exception Handling Demo</title>

</head>

<body>

    <h2>PHP Exception Handling</h2>


    <?php

// Function to divide two numbers safely

function divide($a, $b) {

    if ($b == 0) {
```

```
        throw new Exception("Division by zero is not allowed.");
    }

    return $a / $b;
}

// Function to validate date format (expects YYYY-MM-DD)
function validateDate($date) {

    $d = DateTime::createFromFormat("Y-m-d", $date);

    // Check if parsing failed or format doesn't match exactly
    if (!$d || $d->format("Y-m-d") !== $date) {

        throw new Exception("Invalid date format. Use YYYY-MM-DD.");

    }

    return $d;
}

// Exception handling block
try {

    // Example 1: Division

    $num1 = 10;

    $num2 = 10; // Try changing this to a non-zero value like 2

    $result = divide($num1, $num2);

    echo "Division Result: $num1 / $num2 = $result<br>";

    // Example 2: Date validation

    $dateInput = "2025-03-01"; // Try changing this to a valid date like "2025-09-06"

    $validDate = validateDate($dateInput);

    echo "Valid Date: " . $validDate->format("Y-m-d") . "<br>";
}
```



```
} catch (Exception $e) {  
  
    // Display error message  
  
    echo "<strong>Error:</strong> " . $e->getMessage();  
  
}  
  
?>  
  
</body>  
  
</html>
```